5
SCPI Command Reference

This section contains reference information on programming the Model 2000 with the SCPI commands. It is organized as follows:

SCPI Signal Oriented Measurement Commands — Covers the signal oriented measurement commands. These commands are used to acquire readings.

SCPI command subsystems reference tables — Includes a summary table for each SCPI subsystem command.

SCPI command subsystems — Includes additional information on each SCPI subsystem command.

The signal oriented measurement commands are used to acquire readings. You can use these high-level instructions to control the measurement process. These commands are summarized in Table 5-1.

Table 5-1
Signal oriented measurement command summary

Command	Description
:CONFigure:<function>	Places the Model 2000 in a "one-shot" measurement mode for the specified function.
:FETCh?	Requests the latest reading without triggering.
:READ?	Performs an :ABORt, :INITiate, and a :FETCh?.
MEASure[:<function>]?	Performs an :ABORt, :CONFigure:<function>, and a :READ?.

:CONFigure:<function>

<function> =	CURRent:AC	AC current
	CURRent[:DC]	DC current
	VOLTage:AC	AC voltage
	VOLTage[:DC]	DC voltage
	RESistance	2-wire resistance
	FRESistance	4-wire resistance
	PERiod	Period
	FREQuency	Frequency
	TEMPerature	Temperature
	DIODe	Diode testing
	CONTinuity	Continuity test

Query :CONFigure? Query the selected function.

Description This command configures the instrument for subsequent measurements on the specified function. Basically, this command places the instrument in a "one-shot" measurement mode. You then use the :READ? command to trigger a measurement and acquire a reading (see :READ?).

When this command is sent, the Model 2000 will be configured as follows:

- The function specified by this command is selected.
- All controls related to the selected function are defaulted to the *RST values.
- Continuous initiation is disabled (:INITiate:CONTinuous OFF).
- The control source of the Trigger Model is set to Immediate.
- The count values of the Trigger Model are set to one.

- The delay of the Trigger Model is set to zero.
- The Model 2000 is placed in the idle state.
- All math calculations are disabled.
- Buffer operation is disabled. A storage operation currently in process will be aborted.
- Autozero is set to the *RST default value.
- All operations associated with switching cards (scanning) are disabled.

This command is automatically asserted when the :MEASure? command is sent.

Program

```
PRINT #1, "output 16; :conf:volt:dc"     'Perform :CONFigure operations.
```

FETCh? command

:FETCh?

Description

This query command requests the latest post- processed reading. After sending this command and addressing the Model 2000 to talk, the reading is sent to the computer. This command does not affect the instrument setup.

This command does not trigger a measurement. The command simply requests the last available reading. Note that this command can repeatedly return the same reading. Until there is a new reading, this command continues to return the old reading. If your application requires a "fresh" reading, use the :DATA:FRESh? command (see the SENSe Subsystem command).

This command is automatically asserted when the :READ? or :MEASure? command is sent.

NOTE If external rapid triggers are applied, the unit may not return readings when using :FETCh?

READ? command

:READ?

Description

Typically, this command is used with the instrument in the “one-shot” measurement mode to trigger and acquire a specified number of readings. The :SAMPle:COUNt command is used to specify the number of readings (see Trigger Subsystem). Note that the readings are stored in the buffer.

When this command is sent, the following commands execute in the order that they are presented:

:ABORt
:INITiate
:FETCh?

When :ABORt is executed, if continuous initiation is disabled, the instrument goes into the idle state. If continuous initiation is enabled, the operation re-starts at the beginning of the Trigger Model.

If the instrument is in the idle state, :INITiate takes the instrument out of the idle state. If continuous initiation is enabled, (:INITiate:CONTinuous ON), then the :INITiate command generates an error and ignores the command.

See the :FETCh? command for more details. Note that an “Init ignored” error will not cancel the execution of the :FETCh? command.

NOTE *You cannot use the :READ? command if sample count >1 (see Trigger Subsystem) and there are readings stored in the buffer (error -225, out of memory). Either set sample count to one or clear the buffer.*

See Appendix C for an example program using the READ? command.

MEASure command

:MEASure[:<function>]?

<function> =	CURRent:AC	AC current
	CURRent[:DC]	DC current
	VOLTage:AC	AC voltage
	VOLTage[:DC]	DC voltage
	RESistance	2-wire resistance
	FRESistance	4-wire resistance
	PERiod	Period
	FREQuency	Frequency
	TEMPerature	Temperature
	DIODe	Diode testing
	CONTinuity	Continuity test

Description

This command combines all of the other signal oriented measurement commands to perform a “one-shot” measurement and acquire the reading.

When this command is sent, the following commands execute in the order that they are presented.

:ABORt:CONFigure:<function>:READ?

When :ABORt is executed, if continuous initiation is disabled, the instrument goes into the idle state. If continuous initiation is enabled, the operation re-starts at the beginning of the Trigger Model.

When :CONFigure is executed, the instrument goes into a “one-shot” measurement mode. See :CONFigure for more details.

When :READ? is executed, its operations will then be performed. In general, another :ABORt is performed, then an :INITiate, and finally a FETCh? to acquire the reading. See :READ? for more details.

SCPI command subsystems reference tables

Tables 5-2 through 5-11 summarize the commands for each SCPI subsystem. The following list includes the SCPI subsystem commands and the table number where each command is summarized.

CALCulate command summary (Table 5-2)
DISPlay command summary (Table 5-3)
FORMat command summary (Table 5-4)
ROUTe command summary (Table 5-5)
SENSe command summary (Table 5-6)
STATus command summary (Table 5-7)
SYSTem command summary (Table 5-8)
TRACe command summary (Table 5-9)
Trigger command summary (Table 5-10)
UNIT command summary (Table 5-11)

General notes:

- Brackets ([ ]) are used to denote optional character sets. These optional characters do not have to be included in the program message. Do not use brackets in the program message.
- Angle brackets (< >) are used to indicate parameter type. Do not use angle brackets in the program message.
- The Boolean parameter (<b>) is used to enable or disable an instrument operation. 1 or ON enables the operation and 0 or OFF disables the operation.
- Upper case characters indicate the short-form version for each command word.
- Default Parameter — Listed parameters are both the *RST and :SYSTem:PRESet defaults, unless noted otherwise. Parameter notes are located at the end of each table.
  SCPI — A checkmark (√) indicates that the command and its parameters are SCPI confirmed. An unmarked command indicates that it is non-SCPI. SCPI confirmed commands that use one or more non-SCPI parameters are explained by notes.

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  SCPI — A checkmark (√) indicates that the command and its parameters are SCPI confirmed. An unmarked command indicates that it is non-SCPI. SCPI confirmed commands that use one or more non-SCPI parameters are explained by notes.

Table 5-2
CALCulate command summary

Command	Description	Default parameter	SCPI
:CALCulate[1]	Subsystem to control CALC 1:		√
:FORMat <name>	Select math format (NONE, MXB, PERCent).	PERCent	√
:FORMat?	Query math format.		√
:KMATh	Path to configure math calculations:		√
:MMFactor <NRf>	Set “m” factor for mx+b (-100e6 to 100e6).	1	
:MMFactor?	Query “m” factor.		
:MBFactor <NRf>	Set “b” factor for mx+b (-100e6 to 100e6).	0	
:MBFactor?	Query “b” factor.		
:MUNits <name>	Specify units for mx+b reading (three characters ‘A’ through ‘Z’).	MXB	
:MUNits?	Query “mx+b” units.		
:PERCent <NRf>	Set target value for PERCent calculation (-100e6 to 100e6).	1	
:ACQuire	Use input signal as target value.		
:PERCent?	Query percent.		
:STATe <b>	Enable or disable kmath calculation.	(Note)	
:STATe?	Query state of kmath function.		
:DATA?	Read result of kmath calculation.		
:CALCulate2	Subsystem to control CALC 2:		√
:FORMat <name>	Select math format: (MEAN, SDEViation, MAXimum, MINimum, or NONE).	NONE	√
:FORMat?	Query math format.		√
:STATe <b>	Enable or disable calculation.	(Note)	√
:STATe?	Query state of math function.		√
:IMMediate	Recalculate raw input data in buffer.		√
:IMMediate?	Perform calculation and read result.		√
:DATA?	Read math result of CALC 2.		√
:CALCulate3	Subsystem to control CALC 3 (limit test):		√
:LIMit[1]	Path to control LIMIT 1 test:		√
:UPPer	Path to configure upper limit:		√
[:DATA] <n>	Set upper limit (-100e6 to 100e6).	1	√
[:DATA]?	Query upper limit.		√
:LOWer	Path to configure lower limit:		√
[:DATA] <n>	Set lower limit (-100e6 to 100e6).	-1	√
[:DATA]?	Query lower limit.		√
:STATe <b>	Enable or disable limit test.	OFF	√
:STATe?	Query state of limit test.		√
:FAIL?	Query test result (1 = pass, 0 = fail).		√
:CLEAR	Path to clear failed test:		√
[:IMMediate]	Clear failed test indication.		√
:AUTO <b>	Enable or disable auto clear.	ON	√
:AUTO?	Query auto clear.		√
:IMMediate	Re-perform limit tests.		√

*Note: ON is the *RST default parameter, and OFF is the :SYSTem:PRESet default.

Table 5-3
DISPlay command summary

Command	Description	Default parameter	SCPI
:DISPlay			
[:WINDow[1]]			√
:TEXT	Path to control user text messages.	(Note 1)	√
:DATA <a>	Define ASCII message “a” (up to 12 characters).		√
:DATA?	Query text message.		√
:STATe <b>	Enable or disable message mode.	(Note 2)	√
:STATe?	Query text message state.		√
:ENABle <b>	Enable or disable the front panel display.	(Note 3)	√
:ENABle?	Query state of the display.		√

Notes:
1. *RST and :SYSTem:PRESet has no effect on a user defined message. Cycling power cancels all user defined messages.
2. *RST and :SYSTem:PRESet has no effect on the state of the message mode. Cycling power disables (OFF) the message mode.
3. *RST and :SYSTem:PRESet has no effect on the display circuitry. Cycling power enables (ON) the display circuitry.

Table 5-4
FORMat command summary

Command	Description	Default parameter	SCPI
:FORMat			
[:DATA] <type>[,<length>]	Select data format: (ASCii, SREal or DREal).	ASCii	√
[:DATA]?	Query data format.		√
:ELEMents <item list>	Specify data elements: (READing, CHANnel, and UNITs).	READing	
:ELEMents?	Query data elements.		
:BORDer <name>	Select binary byte order: (NORMal or SWAPped).	SWAPped	√
:BORDer?	Query byte order.		√

Table 5-5
ROUTe command summary

Command	Description	Default parameter	SCPI
ROUTe	Commands to control scanner card:		
:CLOSe <chan num>	Close specified channel (1 to 10) or channel pair (1 to 5).		
:STATe?	Query closed channel (or channel pair).		
:OPEN:ALL	Open all input channels (1 through 10).		
:MULTiple	Path to close and open multiple channels:		
:CLOSe <list>	Close specified channels (1 to 11).		
:STATe?	Query closed channel.		√
:OPEN <list>	Open specified channels (1 to 11).		√
:SCAN	Path to scan channels.		√
[:INTernal] <list>	Specify internal scan list (2 to 10 channels).	1-10	
[:INTernal]?	Query internal scan list.		
:EXTernal <list>	Specify external scan list (2 to 800 channels).	1-10	
:EXTernal?	Query external scan list.		
:LSELect <name>	Select scan operation (INTernal, EXTernal, or NONE).	NONE	
:LSELect?	Query scan operation.		

Table 5-6
SENSe command summary

Command	Description	Default parameter	SCPI
[:SENSe[1]]			
:FUNCtion <name>	Select measurement function: 'VOLTage:AC', VOLTage :DC', RESistance', 'FRESistance', 'CURRent:AC', 'CURRent: DC' , 'FREQuency','TEMPerature', 'PERiod', 'DIODe', "CONTinuity'.	'VOLT[:DC]'	√
:FUNCtion?	Query function.		√
:DATA?	Return the last instrument reading.		√
:HOLD	Path to control Hold feature:		
:WINDow <NRf>	Set Hold window (%); 0.01 to 20.	1	
:WINDow?	Query Hold window.		
:COUNt <NRf>	Set Hold count; 2 to 100.	5	
:COUNt?	Query Hold count.		
:STATe <NRf>	Enable or disable Hold.	OFF	
:STATe?	Query state of Hold.		

Table 5-6(cont.)
SENSe command summary

Command	Description	Default parameter	SCPI
:CURRent:AC	Path to configure AC current.		√
:NPLCycles <n>	Set integration rate (line cycles; 0.01 to 10).	1	√
:NPLCycles?	Query line cycle integration rate.		√
:RANGe	Path to configure measurement range:		√
[:UPPer] <n>	Select range (0 to 3.1).	3	√
[:UPPer]?	Query range.		√
:AUTO <b>	Enable or disable auto range.	ON	√
:AUTO?	Query auto range.		√
:REFerence <n>	Specify reference (-3.1 to 3.1).	0	√
:STATe <b>	Enable or disable reference.	OFF	√
:STATe?	Query state of reference.		√
:ACQuire	Use input signal as reference.		
:REFerence?	Query reference value.		√
:DIGits <n>	Specify measurement resolution (4 to 7).	6	
:DIGits?	Query resolution.		
:AVERage	Path to configure and control the filter.		
:TCONtrol <name>	Select filter type: (MOVing or REPeat).	(Note)	
:TCONtrol?	Query filter type.		
:COUNt <n>	Specify filter count (1 to 100).		
:COUNt?	Query filter count.	10	
:STATe <b>	Enable or disable filter.	OFF	
:STATe?	Query state of digital filter.		
:DETector	Path to configure bandwidth:		
:BANDwidth <NRf>	Specify bandwidth (3 to 300e3).	30	
:BANDwidth?	Query bandwidth.		
:CURRent:DC	Path to configure DC current.		√
:NPLCycles <n>	Set integration rate (line cycles; 0.01 to 10).	1	√
:NPLCycles?	Query line cycle integration rate.		√
:RANGe	Path to configure measurement range:		√
[:UPPer] <n>	Select range (0 to 3.1).	3	√
[:UPPer]?	Query range.		√
:AUTO <b>	Enable or disable auto range.	ON	√
:AUTO?	Query auto range.		√
:REFerence <n>	Specify reference (-3.1 to 3.1).	0	√
:STATe <b>	Enable or disable reference.	OFF	√
:STATe?	Query state of reference.		√
:ACQuire	Use input signal as reference.		
:REFerence?	Query reference value.		√
:DIGits <n>	Specify measurement resolution (4 to 7).	7	
:DIGits?	Query resolution.		
:AVERage	Path to configure and control the filter.		
:TCONtrol <name>	Select filter type: (MOVing or REPeat).	(Note)	
:TCONtrol?	Query filter type.		
:COUNt <n>	Specify filter count (1 to 100).	10	
:COUNt?	Query filter count.		
:STATe <b>	Enable or disable filter.	OFF	
:STATe?	Query state of digital filter.		

Table 5-6(cont.)
SENSe command summary

Command	Description	Default parameter	SCPI
:VOLTage:AC	Path to configure AC voltage.		√
:NPLCycles <n>	Set integration rate (line cycles; 0.01 to 10).	1	√
:NPLCycles?	Query line cycle integration rate.		√
:RANGe	Path to configure measurement range:		√
[:UPPer] <n>	Select range (0 to 757.5).	757.5	√
[:UPPer]?	Query range.		√
:AUTO <b>	Enable or disable auto range.	ON	√
:AUTO?	Query auto range.		√
:REFerence <n>	Specify reference (-757.5 to 757.5).	0	√
:STATe <b>	Enable or disable reference.	OFF	√
:STATe?	Query state of reference.		√
:ACQuire	Use input signal as reference.		
:REFerence?	Query reference value.		√
:DIGits <n>	Specify measurement resolution (4 to 7).	6	
:DIGits?	Query resolution.		
:AVERage	Path to configure and control the filter.		
:TCONtrol <name>	Select filter type: (MOVing or REPeat).	(Note)	
:TCONtrol?	Query filter type.		
:COUNt <n>	Specify filter count (1 to 100).	10	
:COUNt?	Query filter count.		
:STATe <b>	Enable or disable filter.		
:STATe?	Query state of digital filter.	OFF	
:DETector	Path to configure bandwidth:		
:BANDwidth <NRf>	Specify bandwidth (3 to 300e3).	30	
:BANDwidth?	Query bandwidth.		
:VOLTage:DC	Path to configure DC voltage:		√
:NPLCycles <n>	Set integration rate (line cycles; (0.01 to 10).	1	√
:NPLCycles?	Query line cycle integration rate.		√
:RANGe	Path to configure measurement range:		√
[:UPPer] <n>	Select range (0 to 1010).	1000	√
[:UPPer]?	Query range.		√
:AUTO <b>	Enable or disable auto range.	ON	√
:AUTO?	Query auto range.		√
:REFerence <n>	Specify reference (-1010 to +1010).	0	√
:STATe <b>	Enable or disable reference.	OFF	√
:STATe?	Query state of reference (0 or 1).		√
:ACQuire	Use input signal as reference.		
:REFerence?	Query reference value.		√
:DIGits <n>	Specify measurement resolution (4 to 7).	7	
:DIGits?	Query resolution.		
:AVERage	Path to configure and control the filter.		
:TCONtrol <name>	Select filter type: (MOVing or REPeat).	(Note)	
:TCONtrol?	Query filter type.		
:COUNt <n>	Specify filter count (1 to 100).	10	
:COUNt?	Query filter count.		
:STATe <b>	Enable or disable filter.	OFF	
:STATe?	Query state of digital filter.		

Table 5-6(cont.)
SENSe command summary

Command	Description	Default parameter	SCPI
:RESistance	Path to configure resistance:		√
:NPLCycles <n>	Set integration rate (line cycles; 0.01 to 10).	1	√
:NPLCycles?	Query line cycle integration rate.		√
:RANGe	Path to configure measurement range:		√
[:UPPer] <n>	Select range (0 to 120e6).	100e6	√
[:UPPer]?	Query range.		√
:AUTO <b>	Enable or disable auto range.	ON	√
:AUTO?	Query auto range.		√
:REFerence <n>	Specify reference (0 to 120e6).	0	√
:STATe <b>	Enable or disable reference.	OFF	√
:STATe?	Query state of reference.		√
:ACQuire	Use input signal as reference.		√
:REFerence?	Query reference value.		
:DIGits <n>	Specify measurement resolution (4 to 7).	7	
:DIGits?	Query resolution.		
:AVERage	Path to configure and control filter.		
:TCONtrol <name>	Select filter type: (MOVing or REPeat).	(Note)	
:TCONtrol?	Query filter type.		
:COUNt <n>	Specify filter count (1 to 100).	10	
:COUNt?	Query filter count.		
:STATe <b>	Enable or disable filter.	OFF	
:STATe?	Query state of digital filter.		
:FRESistance	Path to configure four-wire resistance:		√
:NPLCycles <n>	Set integration rate (line cycles; 0.01 to 10).	1	√
:NPLCycles?	Query line cycle integration rate.		√
:RANGe	Path to configure measurement range:		√
[:UPPer] <n>	Select range (0 to 101e6).	100e6	√
[:UPPer]?	Query range.		√
:AUTO <b>	Enable or disable auto range.	ON	√
:AUTO?	Query auto range.		√
:REFerence <n>	Specify reference (0 to +101e6).	0	√
:STATe <b>	Enable or disable reference.	OFF	√
:STATe?	Query state of reference.		√
:ACQuire	Use input signal as reference.		√
:REFerence?	Query reference value.		
:DIGits <n>	Specify measurement resolution (4 to 7).	7	
:DIGits?	Query resolution.		
:AVERage	Path to configure and control filter.		
:TCONtrol <name>	Select filter type: (MOVing or REPeat).	(Note)	
:TCONtrol?	Query filter type.		
:COUNt <n>	Specify filter count (1 to 100).	10	
:COUNt?	Query filter count.		
:STATe <b>	Enable or disable filter.	OFF	
:STATe?	Query state of digital filter.		

Table 5-6(cont.)
SENSe command summary

Command	Description	Default parameter	SCPI
:TEMPerature	Path to configure temperature:		
:NPLCycles <n>	Set integration rate (line cycles; 0.01 to 10).	1	
:NPLCycles?	Query line cycle integration rate.		
:REFerence <n>	Specify reference; -200 to 1372	0	
:STATe <b>	Enable or disable reference.	OFF	
:STATe?	Query state of reference.		
:ACQuire	Use input signal as reference.		
:REFerence?	Query reference value.		
:DIGits <n>	Specify measurement resolution (4 to 7).	6	
:DIGits?	Query resolution.		
:AVERage	Path to configure and control the filter.		
:TCONtrol <name>	Select filter type: (MOVing or REPeat).	(Note)	
:TCONtrol?	Query filter type.		
:COUNt <n>	Specify filter count (1 to 100).	10	
:COUNt?	Query filter count.		
:STATe <b>	Enable or disable filter.	OFF	
:STATe?	Query state of digital filter.		
:TCouple	Path to configure thermocouple:		
:TYPE <name>	Select thermocouple type (J, K, or T).	J	
:TYPE?	Query thermocouple type.		
:RJUNction	Path to configure reference junction:		
:RSELect <name>	Select reference type (SIMulated or REAL).	SIMulated	
:RSELect?	Query reference type.		
:SIMulated <n>	Specify simulated temperature in °C (0 to 50).	23°C	
:SIMulated?	Query simulated temperature.		
:REAL	Path to configure real reference junction:		
:TCOefficient <n>	Specify temp coefficient (-0.09999 to 0.09999).	2e-4	
:TCOefficient?	Query temp coefficient.		
:OFFSET <n>	Specify voltage offset at 0°C (-0.09999 to 0.09999).	5.463e-2	
:OFFSet?	Query voltage offset.		
:FREQuency	Path to configure frequency.		
:THReshold	Path to select the threshold voltage range:		
:VOLTage			√
:RANGe <n>	Select threshold range (0 to 1010).		√
:RANGe?	Query threshold range.		
:REFerence <n>	Specify reference (0 to 1.5e7)	10	
:STATe <b>	Enable or disable reference.		
:STATe?	Query state of reference.	0	
:ACQuire	Use input signal as reference.	OFF	
:REFerence?	Query reference value.		
:DIGits <n>	Specify measurement resoltuion (4 to 7).		
:DIGits?	Query resolution.	7	

Table 5-6(cont.)
SENSe command summary

Command	Description	Default parameter	SCPI
:PERiod	Path to configure period.		
:THReshold	Path to select the threshold voltage range:		
:VOLTage			
:RANGe <n>	Select threshold range (0 to 1010).		
:RANGe?	Query threshold range.		
:REFerence <n>	Specify reference (0 to 1).	10	
:STATe <b>	Enable or disable reference.		
:STATe?	Query state of reference.	0	
:ACQuire	Use input signal as reference.	OFF	
:REFerence?	Query reference value.		
:DIGits <n>	Specify measurement resolution (4 to 7).		
:DIGits?	Query resolution.	7	
:DIODe	Paths to configure diode test:		
:CURRent			
:RANGe	Path to select range.		
[:UPPer] <NRf>	Select range (0 to 1e-3)	1e-3	
[:UPPer]?	Query range.		
:CONTinuity	Path to configure continuity test:		
:THReshold <NRf>	Set threshold resistance (1 to 1000).	10	
:THReshold?	Query threshold resistance.		

Note: REPeat is the *RST default and MOVing is the :SYSTem:PRESet default.

Table 5-7
STATus command summary

Command	Description	Default parameter	SCPI
:STATus		(Note 1)	√
:MEASurement	Path to control measurement event registers:		
[:EVENt]?	Read the event register.	(Note 2)	
:ENABle <NRf>	Program the enable register.	(Note 3)	
:ENABle?	Read the enable register.		
:CONDition?	Read the condition register.		√
:OPERation	Path to control operation status registers:		√
[:EVENt]?	Read the event register.	(Note 2)	√
:ENABle <NRf>	Program the enable register.	(Note 3)	√
:ENABle?	Read the enable register.		√
:CONDition?	Read the condition register.		√
:QUEStionable	Path to control questionable status registers:		√
[:EVENt]?	Read the event register.	(Note 2)	√
:ENABle <NRf>	Program the enable register.	(Note 3)	√
:ENABle?	Read the enable register.		√
:CONDition?	Read the condition register.		√
:PRESet	Return status registers to default states.		√
:QUEue	Path to access error queue:		√
[:NEXT]?	Read the most recent error message.	(Note 4)	√
:ENABle <list>	Specify error and status messages for queue.	(Note 5)	√
:ENABle?	Read the enabled messages.		
:DISable <list>	Specify messages not to be placed in queue.	(Note 5)	
:DISable?	Read the disabled messages.		
:CLEar	Clears all messages from Error Queue.		

Notes:
1. Commands in this subsystem are not affected by *RST and :SYSTem:PRESet. The effects of cycling power, *CLS and :STATus:PRESet are explained by the following notes.
2. Event Registers: Power-up and *CLS – Clears all bits of the registers
   :STATus:PRESet – No effect.
3. Enable Registers: Power-up and :STATus:PRESet – Clears all bits of the registers
   *CLS – No effect
4. Error Queue: Power-up and *CLS – Clears the Error Queue
   :STATus:PRESet – No effect
5. Enable/Disable Error Queue Messages: Power-up – Clears list of messages
   *CLS and :STATus:PRESet – No effect

Table 5-8
SYSTem command summary

Command	Description	Default parameter	SCPI
:SYSTem			
:PRESet	Return to :SYST:PRES defaults.		√
:POSetup <name>	Select power-on setup: (RST, PRESet or SAV0).		
:POSetup?	Query power-on setup.		
:FRSWitch?	Query INPUTS switch (0=rear, 1=front).		
:VERSion?	Query rev level of SCPI standard.		√
:ERRor?	Query (read) Error Queue.	(Note)	√
:AZERo	Path to set up autozero.		
:STATe <b>	Enable or disable autozero.	ON	
:STATe?	Query autozero.		
:KEY <NRf>	Simulate key-press (1 to 31; see Figure 5-10).		√
:KEY?	Query the last “pressed” key.		√
:CLEar	Clears messages from the Error Queue.		
:BEEPer	Path to control beeper.		
[:STATe] <b>	Enable or disable beeper.	ON	√
[:STATe]?	Query state of beeper.		√
:LOCal	Take 2000 out of remote and restore operation of front panel controls (RS-232 only).		
:REMote	Place 2000 in remote (RS-232 only).		
:RWLock	Lockout front panel controls (RS-232 only).		
:KCLick <b>	Turn the keyclick on/off.	ON	
:KCLick	Query the keyclick status.		
:LFRequency?	Query power line frequency.		

Note: Clearing the Error Queue: Power-up and *CLS – Clears the Error Queue
*RST, :SYSTem:PRESet and :STATus:PRESet – No effect

Table 5-9
TRACe command summary

Command	Description	Default parameter*	SCPI
:TRACe|:DATA	Use :TRACe or :DATA as root command.		
:CLEar	Clear readings from buffer.		
:FREE?	Query bytes available and bytes in use.		√
:POINts <NRf>	Specify size of buffer (2 to 1024).		√
:POINts?	Query buffer size.		√
:FEED <name>	Select source of readings (SENSe[1], CALCulate[1], NONE.		√
:CONTrol <name>	Select buffer control mode (NEVer or NEXT)		√
:CONTrol?	Query buffer control mode.		√
:FEED?	Query source of readings for buffer.		√
:DATA?	Read all readings in the buffer.		√

*:SYSTem:PRESet and *RST have no effect on the commands in this subsystem.

Table 5-10
Trigger command summary

Command	Description	Default parameter	SCPI
:INITiate	Subsystem command path:		√
[:IMMediate]	Initiate one trigger cycle.		√
:CONTinuous <b>	Enable or disable continuous initiation.	(Note 1)	√
:CONTinuous?	Query continuous initiation.		√
:ABORt	Reset trigger system.		√
:TRIGger[:SEQuence[1]]	Path to program Trigger Layer:		√
:COUNt <n>	Set measure count (1 to 9999, or INF).	(Note 2)	√
:COUNt?	Query measure count.		√
:DELay <n>	Set delay (0 to 999999.999 sec)	0	√
:AUTO <b>	Enable or disable auto delay.	OFF	
:AUTO?	Query state of delay.		
:DELay?	Query delay.		
:SOURce <name>	Select control source (IMMediate, TIMer, MANual, BUS, or EXTernal).	IMMediate	√
:SOURce?	Query control source.		
:TIMer <n>	Set timer interval (0.001 to 999999.999 sec).	0.1	√
:TIMer?	Request the programmed timer interval.		√
:SIGNal	Loop around control source.		√
:SAMPle			√
:COUNt <NRf>	Specify sample count (1 to 1024).	1	
:COUNt?	Query sample count.		

Notes:
1. Defaults for continuous initiation:
   :SYSTem:PRESet enables continuous initiation
   *RST disables continuous initiation
2. Defaults for count:
   :SYSTem:PRESet sets the count to INF (infinite).
   *RST sets the count to 1.

Table 5-11
UNIT command summary

Command	Description	Default parameter	SCPI
:UNIT			
:TEMPerature <name>	Select temperature measurement units (C, F, or K).	C	√
:TEMPerature?	Query temperature units.		√
:VOLTage	Path to configure voltage units.		
:AC <name>	Select ACV measurement units (V, DB or DBM).	V	
:DB	Path to set DB reference voltage.		
:REFerence <n>	Specify reference in volts (le-7 to 1000).	1	
:REFerence?	Query DB reference.		
:DBM	Path to set DBM reference impedance.		
:IMPedance <n>	Specify reference impedance (1 to 9999).	75	
:IMPedance?	Query DBM reference impedance.		
:AC?	Query ACV units.		
[:DC] <name>	Select DCV measurement units (V, DB, or DBM)	V	
:DB	Path to set DB reference voltage:		
:REFerence <n>	Specify reference in volts (1e-7 to 1000).	1	
:REFerence?	Query reference.		
:DBM	Path to set DBM refernece impedance:		
:IMPedance <n>	Specify reference impedance (1 to 9999).	75	
:IMPedance?	Query reference impedance.		
:DC?	Query DCV units.		